



Studybeesofficial

Where Minds Pollinate

Class 10 Biology

Chapter 4

Short Questions Notes

NBF book

Follow Our Socials



studybeesofficial



studybeesofficialpak



studybeesofficialpak

URINARY SYSTEM

Extensive Answer Questions

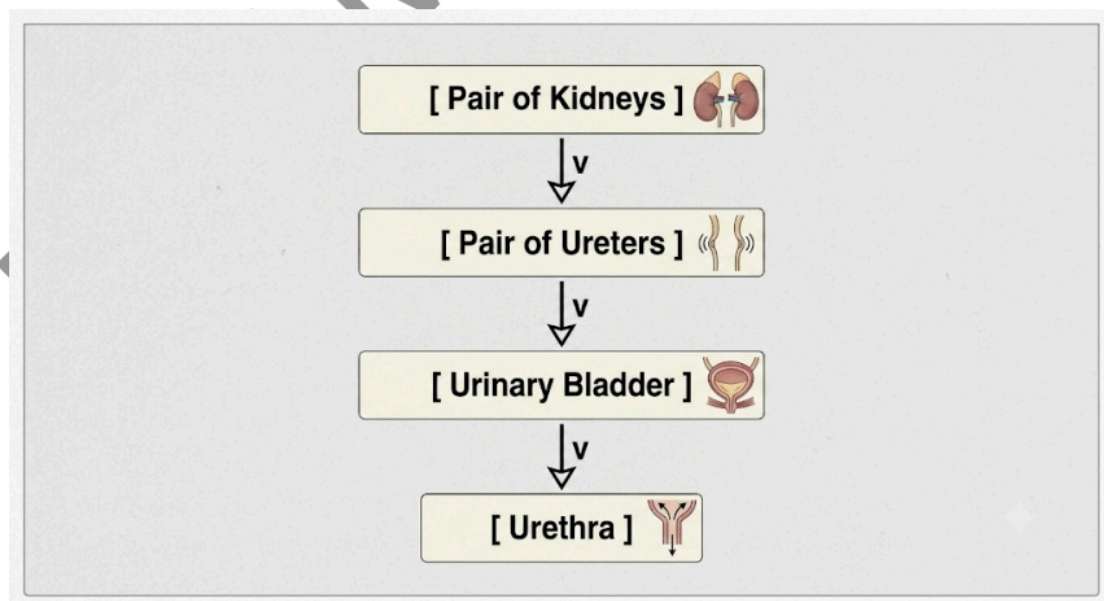
(10th Class Biology)

1. Describe the human urinary system.

The human urinary system (excretory system) is responsible for filtering metabolic waste products from the blood and removing them from the body in the form of urine. It plays a critical role in maintaining water, electrolyte, and pH balance in the body.

Components of the Human Urinary System

The human urinary system consists of four main parts:



1. Pair of Kidneys:

- **Structure:** Dark red, bean-shaped organs located against the back wall of the abdominal cavity on either side of the vertebral column, just below the diaphragm.
- **Function:** Filter blood to remove nitrogenous wastes (mainly urea) and regulate water balance.

2. Pair of Ureters:

- **Structure:** Muscular tubes that emerge from the hilum of each kidney.
- **Function:** Transport urine continuously from the kidneys to the urinary bladder by peristaltic movements.

3. Urinary Bladder:

- **Structure:** A smooth, collapsible, muscular sac situated in the pelvic cavity.
- **Function:** Temporarily stores urine until it is excreted from the body.

4. Urethra:

- **Structure:** A canal that leads from the floor of the urinary bladder to the exterior of the body.
- **Function:** Transports urine out of the body. In males, it also serves as a passage for semen, whereas in females, it functions solely for urine discharge.

2. Relate the structure of the kidney with its function.

The gross anatomical structure of the kidney is uniquely adapted to perform its primary function of blood filtration, urine formation, and excretion.

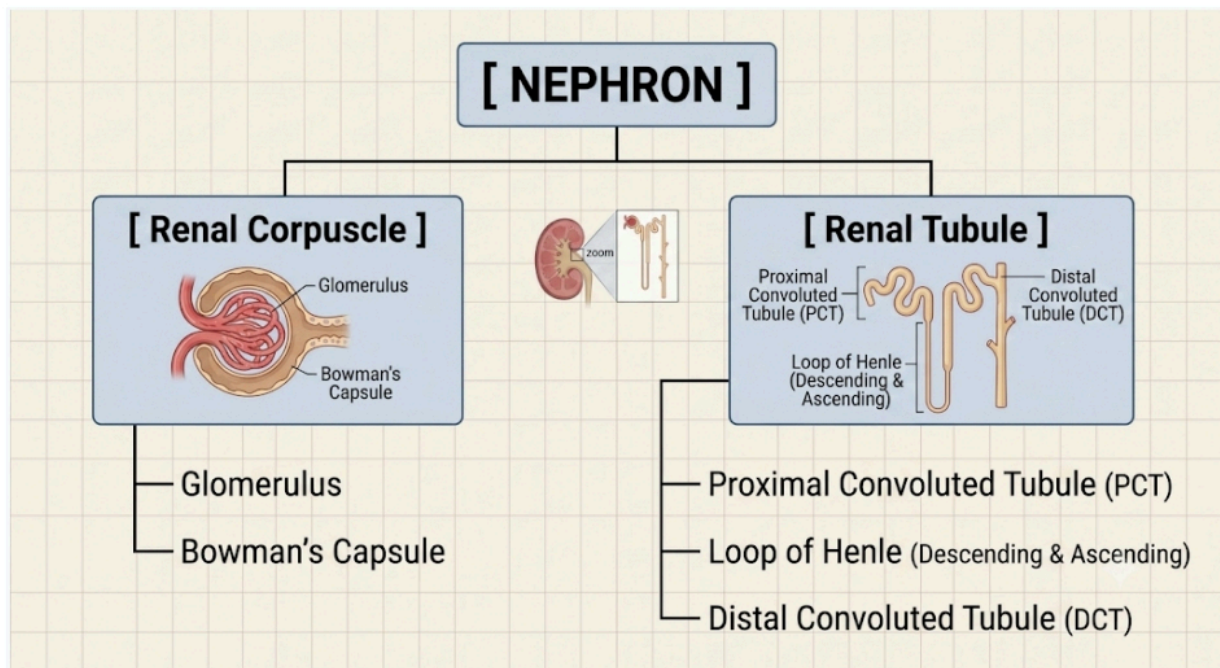
Structure-Function Relationships of the Kidney

Anatomical Structure	Structural Description	Primary Function
Renal Capsule	Tough, fibrous outer membrane surrounding the kidney.	Protects inner kidney tissues from physical trauma and infection.
Renal Cortex	Darker outer region	Serves as the site for

Anatomical Structure	Structural Description	Primary Function
	containing glomeruli and convoluted tubules.	ultrafiltration and selective reabsorption.
Renal Medulla	Lighter inner region composed of 8–18 cone-shaped Renal Pyramids .	Contains loops of Henle and collecting ducts to concentrate urine.
Renal Pyramids	Striated triangular structures within the medulla.	Collect urine formed by nephrons and direct it toward the renal papilla.
Renal Pelvis	Funnel-shaped cavity at the hilum of the kidney.	Acts as a central basin to collect urine before it enters the ureter.
Hilum	A notch on the concave inner border of the kidney.	Serves as the point of entry and exit for the renal artery, renal vein, nerves, and ureter.

3. Describe the structure of the nephron.

The **nephron** is the structural and functional unit of the kidney. Each human kidney contains approximately 1 million nephrons. A nephron consists of two major parts: the **Renal Corpuscle** and the **Renal Tubule**.



1. Renal Corpuscle

The initial filtering component located in the renal cortex, comprising two parts:

- **Glomerulus:** A dense network of high-pressure capillaries branching from the afferent arteriole.
- **Bowman's Capsule:** A cup-shaped double-walled sac surrounding the glomerulus that receives the glomerular filtrate.

2. Renal Tubule

A long, convoluted tube continuous with Bowman's capsule, divided into three main regions:

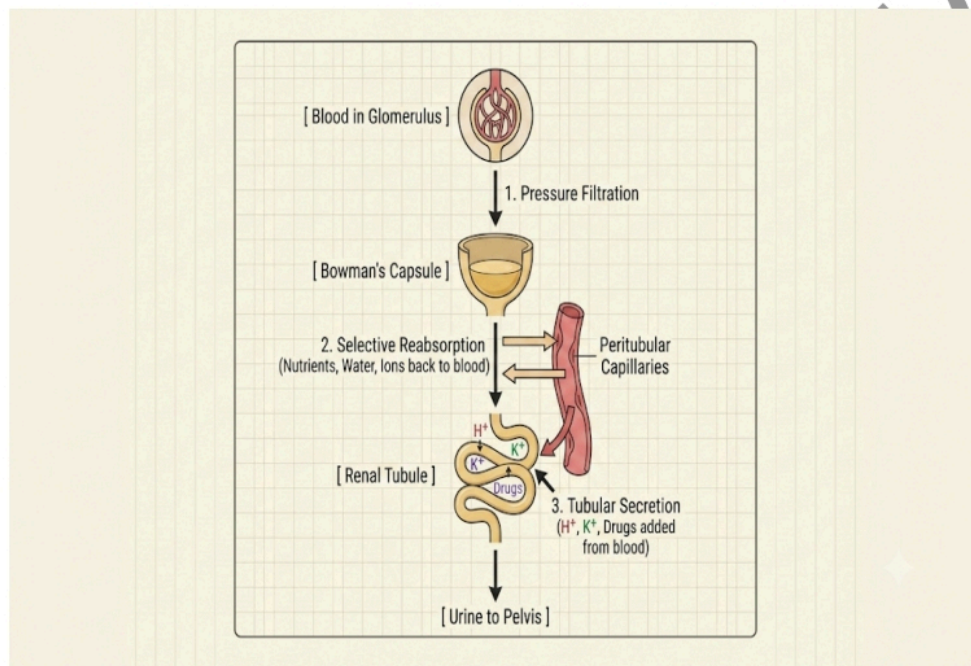
- **Proximal Convoluted Tubule (PCT):** Highly coiled segment in the cortex where most selective reabsorption of glucose, amino acids, and water occurs.
- **Loop of Henle:** A U-shaped loop extending down into the renal medulla, consisting of:
 - *Descending Limb:* Permeable to water, impermeable to salts.
 - *Ascending Limb:* Permeable to salts (Na^+ , Cl^-), impermeable to water.
- **Distal Convoluted Tubule (DCT):** Coiled tubule in the cortex that performs tubular secretion and fine-tunes electrolyte/water balance under hormonal control.

- **Collecting Duct:** Receives urine from multiple nephrons and carries it through the medulla to the renal pelvis.

4. Explain the process of urine formation in humans.

Urine formation involves three distinct physiological processes within the nephron:

Urine Formation = Pressure Filtration - Selective Reabsorption + Tubular Secretion



Phase 1: Pressure Filtration (Ultrafiltration)

- Occurs at the **Renal Corpuscle**.
- High blood pressure in the glomerulus forces water, glucose, amino acids, urea, uric acid, and salts through the capillary walls into Bowman's capsule.
- Blood cells and large plasma proteins are retained in the blood capillaries.
- The resulting fluid in Bowman's capsule is called **Glomerular Filtrate**.

Phase 2: Selective Reabsorption

- Occurs as the filtrate moves through the **Renal Tubule**.
- **In PCT:** About 99% of useful substances (all glucose, amino acids, and most Na^+

and H_2O) are reabsorbed back into the peritubular capillaries by active transport and osmosis.

- **In Loop of Henle:** Water is reabsorbed in the descending limb; salts (Na^+ , Cl^-) are reabsorbed in the ascending limb.
- **In DCT & Collecting Duct:** Additional water is reabsorbed under the influence of Antidiuretic Hormone (ADH).

Phase 3: Tubular Secretion

- Occurs along the **DCT** and **Collecting Duct**.
- Epithelial cells of the tubules actively secrete waste ions (H^+ , K^+ , NH_4^+), urea, and drug residues from peritubular blood into the tubular fluid.
- Maintains blood pH within the normal range (7.35–7.45).

The final fluid remaining in the collecting ducts is **urine**, which drains into the renal pelvis.

5. Describe the osmoregulatory function of kidney.

Osmoregulation is the process of regulating the osmotic pressure of blood and tissue fluids by controlling water and salt balance in the body. The kidney acts as the primary osmoregulatory organ under hormonal control.

Mechanism of Osmoregulation

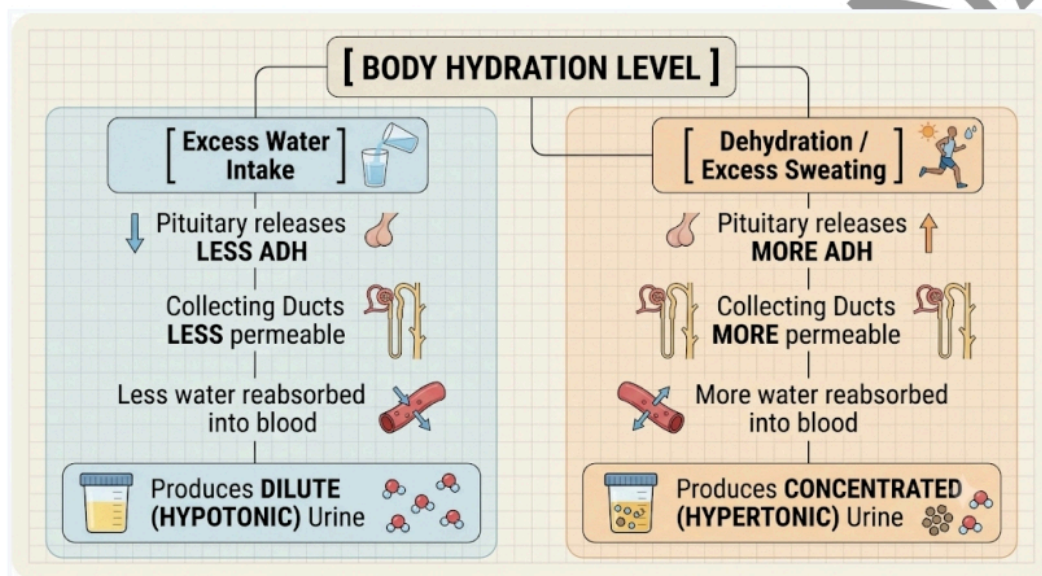
The kidney adjusts urine volume and concentration depending on the body's hydration state:

1. Hypotonic Environment (Excess Body Water)

- **Condition:** High water intake leads to diluted blood plasma.
- **Response:** The pituitary gland decreases the release of Antidiuretic Hormone (ADH).
- **Effect:** Walls of the collecting ducts become less permeable to water, reducing reabsorption into the blood.
- **Result:** Large volume of **dilute (hypotonic) urine** is produced to eliminate excess water.

2. Hypertonic Environment (Water Deficit / Dehydration)

- **Condition:** Profuse sweating, low intake, or high salt intake concentrates blood plasma.
- **Response:** The pituitary gland secretes increased levels of ADH into the bloodstream.
- **Effect:** ADH increases the permeability of the distal convoluted tubule and collecting ducts to water.
- **Result:** Maximum water is reabsorbed back into the blood, producing a small volume of **concentrated (hypertonic) urine**.



6. Discuss kidney stones and their treatment.

Kidney stones (**Nephrolithiasis**) are hard, mineral deposits that precipitate out of urine and form within the renal pelvis or calyces.

Causes and Composition

- **Chemical Composition:**
 - Calcium Oxalate stones (most common, ~70%)
 - Calcium Phosphate stones
 - Uric Acid stones
 - Cystine/Struvite stones
- **Risk Factors:** Low water intake, high dietary oxalate or salt consumption,

hypercalcemia, urinary tract infections (UTIs), and metabolic disorders.

Symptoms

- Severe, sharp pain in the flank (side/back) radiating to the lower abdomen/groin (renal colic).
- Hematuria (blood in urine).
- Painful or frequent urination.
- Nausea and vomiting.

Treatment Methods

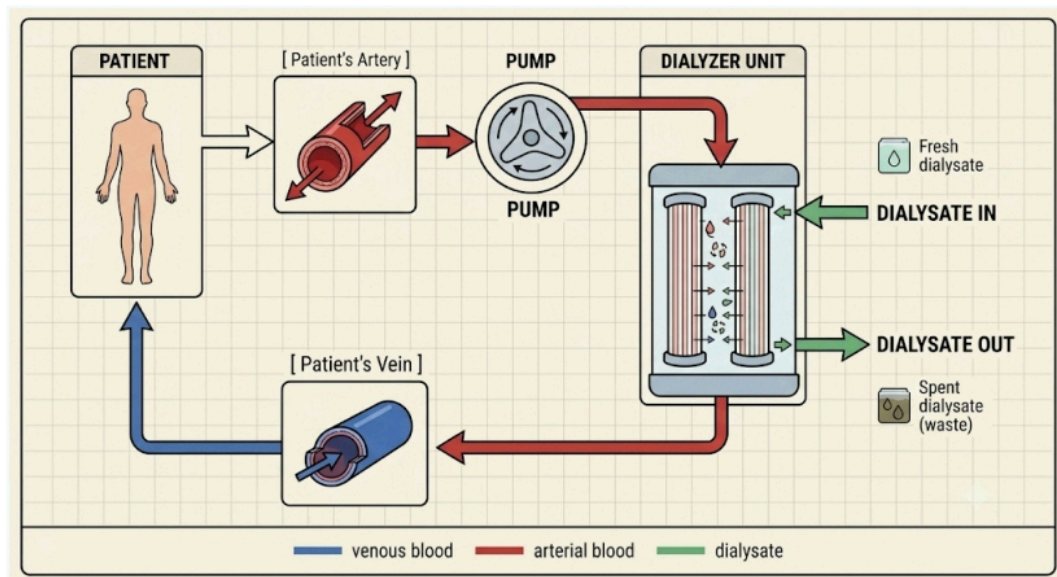
1. **Increased Hydration:**
 - **For small stones (< 5 mm):** Drinking 2–3 liters of water daily allows stones to pass naturally through the urinary tract.
2. **Lithotripsy (Extracorporeal Shock Wave Lithotripsy - ESWL):**
 - **Mechanism:** Non-invasive procedure where high-energy acoustic shock waves are focused from outside the body onto the stone.
 - **Effect:** Breaks the kidney stone into tiny sand-like particles that pass out easily with urine.
3. **Surgical Removal:**
 - **Percutaneous Nephrolithotomy (PCNL) / Ureteroscopy:** Used for larger or complex stones that cannot be broken by shock waves. The stone is removed using an endoscope or minor surgical incision.

7. Describe the following two types of Dialysis

Dialysis is an artificial process of filtering nitrogenous wastes, excess salts, and water from the blood when the kidneys fail to function.

a. Haemodialysis

In hemodialysis, blood is cleaned outside the body using a machine called a **dialyzer** (artificial kidney)

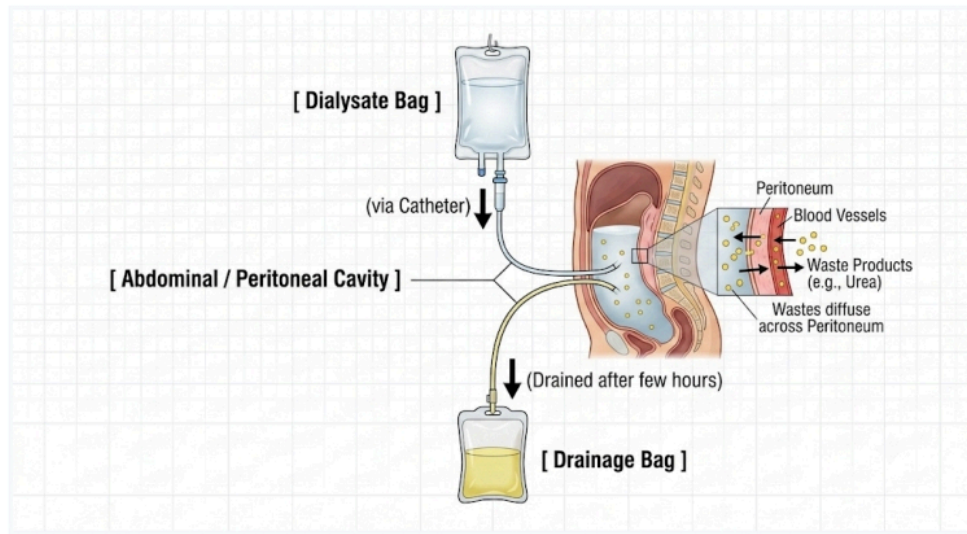


Procedure & Working Mechanism

1. **Vascular Access:** Blood is drawn from an artery (via an AV fistula) and pumped into the dialyzer.
2. **Filtering Unit:** The dialyzer contains thousands of thin, hollow semi-permeable cellophane tubes submerged in **dialysis fluid (dialysate)**.
3. **Diffusion Process:**
 - Dialysate contains no nitrogenous wastes, so urea, uric acid, and excess salts diffuse across the semi-permeable membrane from the blood into the dialysate.
 - Red blood cells, proteins, and essential components stay in the blood.
4. **Return:** Cleansed blood passes through a bubble trap and returns to the patient through a vein.
5. **Frequency:** Usually performed 3 times a week for 3–4 hours per session at a clinic or hospital.

b. Peritoneal Dialysis

In peritoneal dialysis, the natural membrane lining the patient's abdominal cavity (**peritoneum**) acts as the semi-permeable filtering membrane.



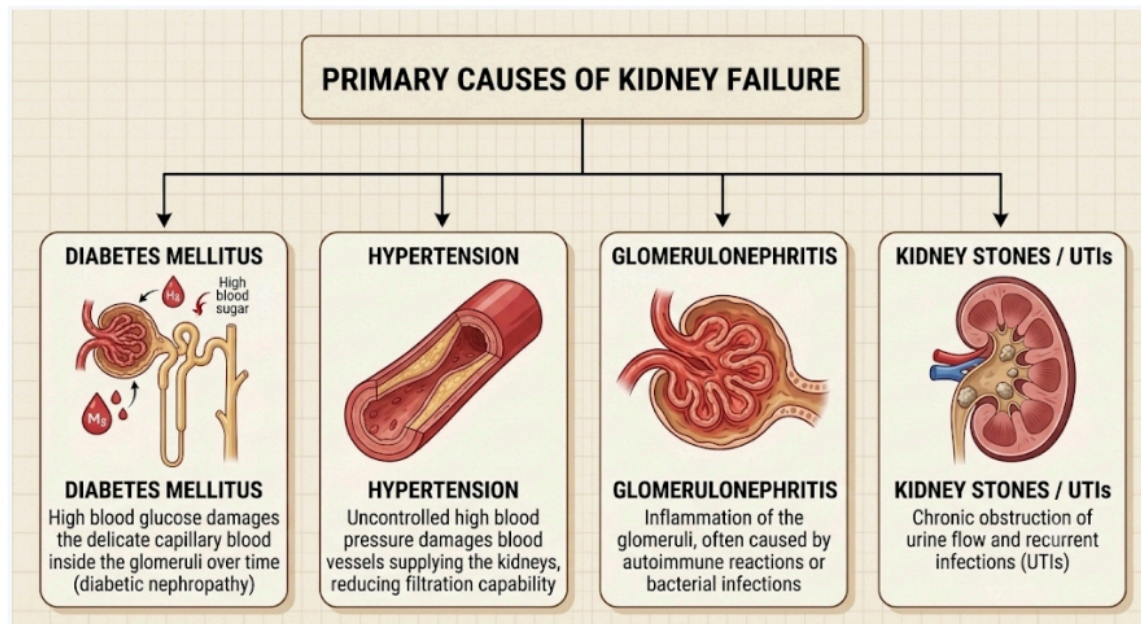
Procedure & Working Mechanism

1. **Catheter Placement:** A flexible tube (catheter) is surgically implanted into the lower abdominal wall.
2. **Infusion:** Dialysis fluid is run into the peritoneal cavity through the catheter.
3. **Dwell Time:** Fluid remains in the cavity for several hours (dwell phase). Metabolic wastes and excess fluid diffuse across the blood capillaries of the peritoneum into the dialysate solution.
4. **Drainage:** Used fluid containing waste products is drained out into a waste bag and replaced with fresh dialysate.
5. **Frequency:** Can be done at home multiple times daily (Continuous Ambulatory Peritoneal Dialysis - CAPD) or overnight using an automated machine.

8. Outline the causes of kidney failure and its treatment.

Kidney failure (**Renal Failure**) is the complete or partial inability of the kidneys to filter metabolic waste products and balance fluids in the body.

Causes of Kidney Failure



1. **Diabetes Mellitus:** High blood glucose damages the delicate capillary blood vessels inside the glomeruli over time (diabetic nephropathy).
2. **Hypertension (High Blood Pressure):** Uncontrolled high blood pressure damages blood vessels supplying the kidneys, reducing filtration capability.
3. **Glomerulonephritis:** Inflammation of the glomeruli, often caused by autoimmune reactions or bacterial infections.
4. **Chronic Obstructive Uropathy:** Long-term obstruction of urine flow due to large kidney stones or prostate enlargement.
5. **Nephrotoxic Substances:** Prolonged misuse of nonsteroidal anti-inflammatory drugs (NSAIDs), severe chemical poisoning, or heavy alcohol abuse.

Treatments for Kidney Failure

1. Dialysis (Temporary Management)

Provides artificial filtration to remove wastes and maintain fluid levels when kidneys cannot function.

- **Hemodialysis:** Machine-based external blood filtration.
- **Peritoneal Dialysis:** Internal filtration using the abdominal lining.

2. Kidney Transplantation (Permanent Solution)

- **Procedure:** Surgical replacement of a damaged kidney with a healthy donor

kidney from a living or deceased donor.

- **Tissue Matching:** The donor's Human Leukocyte Antigens (HLA) and blood group must closely match the recipient to prevent immune rejection.
- **Post-Transplant Care:** The recipient must take **immunosuppressive drugs** for life to prevent the immune system from attacking and rejecting the donor organ.

MA'AM SHABANA

BIOLOGY

CLASS 10TH

CH: 4 URINARY SYSTEM

Short Answer Questions

Q1. What is the role of the urinary system in the body?

Answer: The urinary system removes waste products from the blood and maintains the balance of water, salts, and other important substances. It helps in keeping the body clean and balanced.

Q2. Name the structures of the urinary system.

Answer: The main structures of the urinary system are:

- Kidneys
- Ureters
- Urinary bladder
- Urethra

Q3. Write the functions of the following:

- a) **Kidney:** Removes waste, excess salts, and water from the blood and forms urine.
- b) **Ureter:** Transports urine from the kidney to the urinary bladder.
- c) **Urinary bladder:** Stores urine temporarily.
- d) **Urethra:** Conducts urine out of the body.
- e) **Pelvis of the kidney:** Collects urine before it passes into the ureter.
- f) **Capsule of the kidney:** Protects the kidney from injury and infection.
- g) **Dialysis:** An artificial method to remove waste from blood when the kidneys fail.

Q4. What materials are returned to the blood?

Answer: Useful materials like glucose, amino acids, most water, and some salts are returned to the blood during reabsorption in the kidneys.

Q5. What materials leave the body in urine?

Answer: Urea, extra salts, excess water, and other nitrogenous wastes leave the body in urine.

Q6. Name the disorders of the kidney.

Answer: Kidney stones, Kidney failure, Urinary tract infection (UTI), Nephritis

Q7. Write the causes of the formation of a kidney stone.

Answer:

- Drinking less water (dehydration)
- Excess salt or calcium in urine
- Infection in the urinary tract
- Eating too much protein or salt

Q8. Name the structures in the urinary system of man that are associated with each of these:

a) Urea formation: Liver (but removed by kidneys)

b) Urine formation: Kidney (nephrons)

c) Filtration: Glomerulus in the nephron

d) Reabsorption: Tubules of the nephron

e) Tubular secretion: Tubules of the nephron

f) Temporary storage of urine: Urinary bladder

g) Conduction of urine out of the body: Urethra

Q9. Diagram question (nephron blood pathway)

: Renal artery → afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries → renal vein

Q10. Describe the kidney transplant.

Answer: A kidney transplant is a surgical operation in which a damaged kidney is replaced with a healthy kidney from a donor. It helps the patient live a normal life.

Q11. What is kidney failure and dialysis?

Answer: **Kidney failure:** When the kidneys stop working properly and cannot remove wastes from the blood. **Dialysis:** A medical treatment that removes waste materials and extra water from the blood with the help of a machine.

Extensive Answer Questions

Q1. Describe the human urinary system.

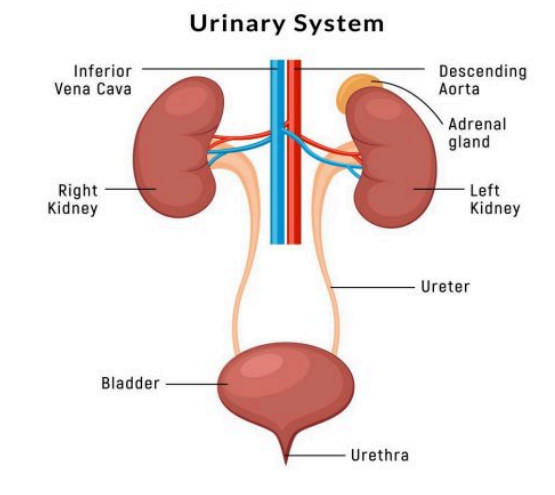
The human urinary system is one of the most vital excretory systems in the body, as it plays a central role in maintaining internal balance and overall health. Its primary function is to filter nitrogenous wastes and other harmful substances from the blood while also regulating the body's water and salt levels, which are crucial for maintaining homeostasis. This system is made up of four main organs: **the kidneys, ureters, urinary bladder, and urethra**, all of which work together in a highly coordinated manner.

The kidneys, which are two bean-shaped organs located on either side of the backbone in the abdominal cavity, serve as the body's natural filters. They not only remove wastes like urea, creatinine, and toxins but also regulate blood pressure, control the levels of electrolytes such as sodium and potassium, and maintain the acid-base balance of the blood. Each kidney contains millions of microscopic units called nephrons, which are the actual filtering structures.

Once the kidneys produce urine, it travels through narrow, muscular tubes called ureters. These ureters transport urine from the kidneys to the urinary bladder with the help of wave-like muscular movements known as peristalsis. The urinary bladder acts as a temporary storage organ, capable of expanding like a balloon to hold urine until the body is ready to expel it. When the bladder becomes full, stretch receptors send signals to the brain, creating the urge to urinate.

Finally, urine exits the body through the urethra, a small tube that carries the urine outside during the process of urination. This process ensures that harmful substances are continuously removed from the body, preventing their accumulation and protecting organs from damage.

In short, the urinary system is not just about waste removal but also about maintaining the body's chemical stability.



Q2. Relate the structure of the kidney to its function.

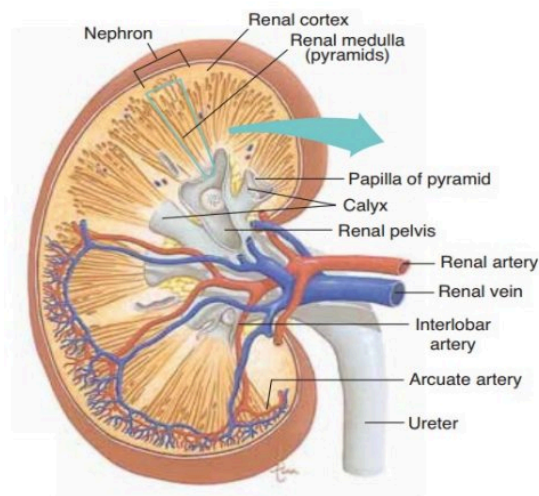
Answer: The kidney is a bean-shaped organ, and its internal structure is closely related to its functions. The kidney has three main regions:

Cortex (outer region): It contains nephrons, which are the functional units of the kidneys. The nephrons filter blood and remove waste products.

Medulla (middle region): It has renal pyramids and collecting ducts. These structures help in concentrating urine and transporting it towards the renal pelvis.

Renal pelvis (inner region): This is a funnel-shaped cavity that collects urine and passes it to the ureter.

The kidney is covered by a protective capsule. Its structure is designed in such a way that it filters about 180 litres of blood daily, reabsorbs useful substances like glucose and salts, and excretes harmful substances as urine.

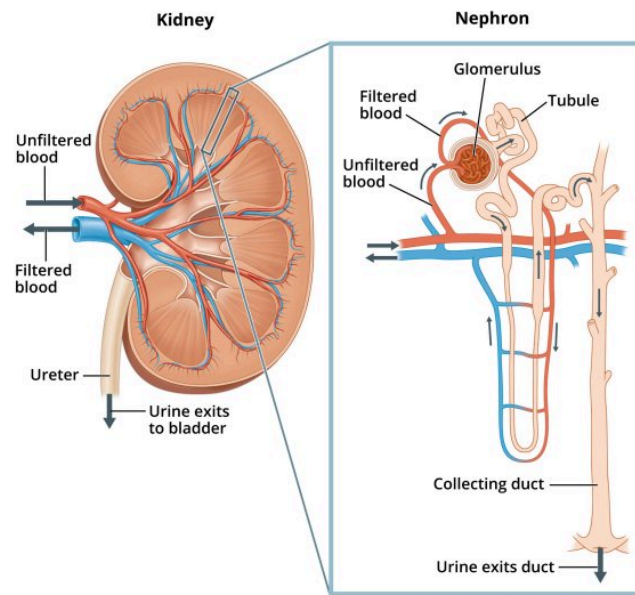


Q3. Describe the structure of a nephron.

Answer: The nephron is the functional unit of the kidney, and each kidney has about 1 million nephrons. A nephron has the following parts:

- **Bowman's capsule:** A cup-like structure that surrounds a cluster of capillaries called the glomerulus. It performs the filtration of blood.
- **Proximal convoluted tubule (PCT):** Lies after the Bowman's capsule. It reabsorbs water, glucose, amino acids, and salts back into the blood.
- **Loop of Henle:** A U-shaped structure important for water and salt balance. It makes the urine more concentrated.
- **Distal convoluted tubule (DCT):** Helps in the secretion of additional wastes and regulates salts.
- **Collecting duct:** Collects urine from many nephrons and passes it towards the renal pelvis.

Thus, the nephron is specially structured to filter blood, reabsorb useful substances, and remove waste products.



Q4. Explain the process of urine formation in humans.

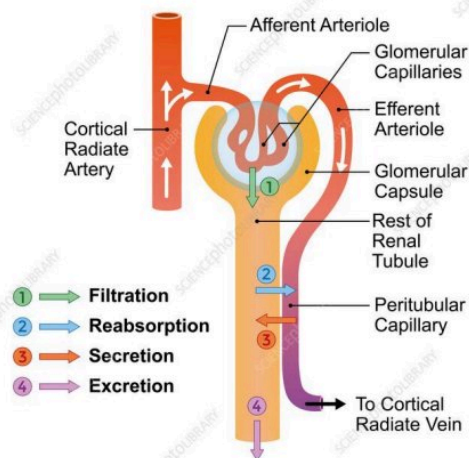
Answer: The formation of urine in humans takes place in three main steps inside the nephrons:

Filtration: Blood enters the glomerulus under high pressure. Wastes, water, salts, glucose, and amino acids pass into Bowman's capsule. This filtrate is called the glomerular filtrate.

Reabsorption: In the proximal tubule and loop of Henle, useful substances like glucose, amino acids, and most of the water are reabsorbed into the blood.

Secretion: In the distal tubule, additional wastes like hydrogen ions, potassium ions, and drugs are secreted into the tubule.

The final fluid left behind is urine, which contains urea, extra salts, and water. It passes to the collecting duct, then to the bladder, and finally out of the body.



Q5. Describe the osmoregulatory function of the kidney.

Answer: Osmoregulation is the process of maintaining the balance of water and salts in the body, primarily through the kidneys. When there's excess water, the kidneys excrete diluted urine; when dehydrated, they reabsorb more water and produce concentrated urine. This balance is controlled by a hormone called ADH (antidiuretic hormone). If water is less in the body, ADH increases, and the kidneys absorb more water. If water is more, ADH decreases, and more water is excreted. Thus, the kidney maintains the right balance of water and salts in the blood, which is necessary for survival.

Q6. Discuss kidney stones and their treatment.

Answer: Kidney stones are hard deposits formed from salts, calcium, or uric acid in the kidneys. They block the urinary tract and cause pain.

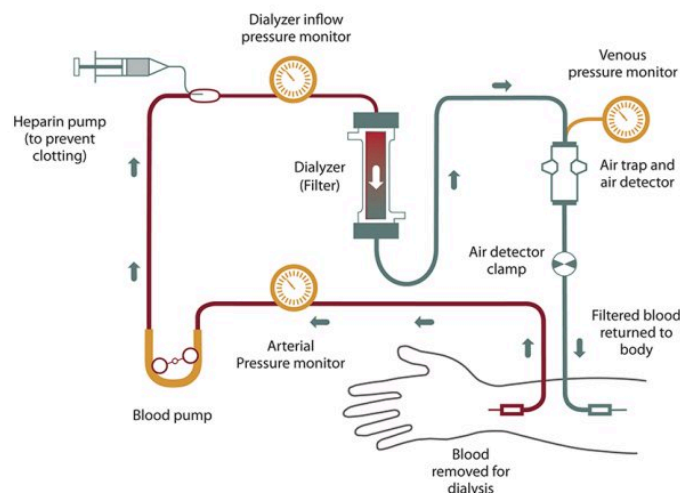
Causes: Drinking too little water, eating too much protein or salt, infections, and high calcium in urine.

Symptoms: Severe back or side pain, blood in urine, nausea, and frequent urination.

Treatment: Small stones may pass out naturally by drinking more water. Medicines can be given to dissolve or pass stones. For larger stones, lithotripsy (using sound waves to break stones) or surgery may be done. Preventive measures include drinking plenty of water, eating a balanced diet, and avoiding excess salt.

Q7. Describe the following two types of Dialysis:

a) Haemodialysis: In haemodialysis, the patient's blood is taken out of the body and passed through a dialysis machine. The machine filters out waste products and extra salts using a special membrane. The clean blood is then returned to the patient's body. This process is done several times a week in hospitals.



b) Peritoneal Dialysis: In this method, a special dialysis fluid is put into the patient's abdominal cavity. The lining of the abdomen (peritoneum) acts as a natural filter. Waste products pass into the fluid, which is later removed and replaced. This method can be done at home.

Both types of dialysis help patients with kidney failure survive when their kidneys cannot work.

Q8. Outline the causes of kidney failure and its treatment.

Answer: Causes of Kidney Failure:

- Long-term diseases like diabetes and high blood pressure.
- Severe kidney infections.
- Kidney stones are blocking urine flow.
- Use of harmful medicines over a long time.
- Polycystic kidney disease (genetic).

Treatment:

Dialysis: Removes wastes artificially when the kidneys fail.

Kidney transplant: A damaged kidney is replaced with a healthy donor kidney.

Medicines and a healthy lifestyle: Control of blood pressure, diabetes, and avoiding harmful drugs.

Early diagnosis and proper treatment can save patients from complete kidney failure.

MA'AM SHABANA